

Aakarsh Arora

Backend Software Engineer — Golang — Distributed Systems — Fintech

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GitHub — LinkedIn — LeetCode — Portfolio

Summary

Backend Engineer with 1.5+ years building production-grade microservices in Golang and Node.js for high-throughput fintech workloads. Specialize in idempotent payment systems, distributed concurrency, and asynchronous processing pipelines integrating Stripe, Coinbase, and 4+ global payment providers. Recently extending backend systems with LLM-powered workflows (RAG, hybrid retrieval) as side projects.

Skills

Languages: Golang, JavaScript (Node.js), Python, Java, SQL

Backend: Microservices, REST, gRPC, Middleware, BFF Pattern, Webhook Handling, Rate Limiting

Distributed Systems: Concurrency, Idempotency, Event-Driven Architecture, Fault Tolerance, Reconciliation

Async & Messaging: Redis Streams, PubSub, Message Queues, Background Workers

Databases: Redis, Aerospike, MongoDB, MySQL, PostgreSQL

Cloud & DevOps: AWS (EC2, Lambda, S3, API Gateway), Docker, Kubernetes, CI/CD

AI / LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Retrieval, LangGraph, Vector Search

Concepts: System Design, Caching, Data Consistency, Performance Optimization, Observability

Experience

Backend Software Engineer, FanCraze (Web3 Fintech / Sports NFTs) — Mumbai *Nov 2024 – Present*

- Migrated **KYC microservice** from Node.js to Golang, reducing verification completion time by **70%** and improving onboarding experience for Indian/International users.
- Migrated **Finance microservice** from Node.js to Golang, improving concurrency and reliability for transaction workflows processing **10K+ requests/day**.
- Designed **idempotent payment processing** with deterministic request keys and database-level constraints, eliminating duplicate charges across distributed retries and webhook replays.
- Integrated **6 payment gateways** (Stripe, Coinbase, Tazapay, Mesh, Mudrex, Shift4) with a unified abstraction layer, fault-tolerant retry logic, and webhook signature verification for secure transaction flows.
- Built **real-time alerting pipelines** notifying the ops team on payment failures, KYC drops, and reconciliation drift — enabling sub-minute incident response and reducing manual monitoring overhead.
- Built **asynchronous processing pipelines** using Redis Streams and message queues, decoupling user-facing APIs from heavy background jobs and improving system resilience under load.
- Architected backend for a **low-latency cricket trading platform** (Uniswap-style blockchain integration), designing caching strategies and DB schema decisions that reduced hot-path API latency by **30%**.
- Implemented **financial reconciliation systems** comparing internal ledger state against payment-provider records, surfacing data drift and ensuring cross-service consistency.
- Led backend delivery for the **core homepage feature**, owning API contracts end-to-end across frontend, admin, and product teams.
- Designed **BFF (Backend-for-Frontend)** services to aggregate downstream microservice calls, reducing client round-trips and payload size.
- Leveraged **Redis and Aerospike** for sub-millisecond reads on high-throughput endpoints serving user balances, KYC state, and trading data.

Software Engineering Intern, DesiQna — Remote

Jun 2023 – Aug 2023

- Built REST APIs in Node.js with modular service architecture.
- Implemented **JWT-based authentication** with Redis-backed session handling and token revocation flows.

Projects

Payment Orchestration Service — Golang, Redis, Message Queues

Scalable payment orchestration engine modeling real-world gateway routing. Implemented **idempotency keys**, exponential-backoff retries, dead-letter queues, multi-provider failover, and per-merchant rate limiting. Designed to handle concurrent transaction bursts without double-charging or data drift.

CodeLeela — *Node.js, Express, MongoDB, Redis*

[\[Live\]](#) — [\[Repo\]](#)

Backend-focused social platform with personalized feeds, JWT authentication, and Redis-cached read paths. Designed REST API layer and Mongo schemas to support post creation, feed ranking, and user interactions.

Video Dating MVP — *Supabase, PostgreSQL, Realtime APIs*

Shipped end-to-end MVP in 1 week: designed Postgres schema, real-time matching workflows, auth, and media handling on Supabase. Validated product feasibility for video-first matching with internal user testing.

Education

B.Tech in Computer Science, Shri Ramswaroop Memorial University

2020 – 2024

Achievements & Leadership

- Solved **400+ DSA problems** on LeetCode; strong foundation in algorithms, data structures, and system design.
- **Led college dramatics team** of 20+ members across 10+ productions, winning 7+ inter-college competitions — directed rehearsals, coordinated cross-functional teams (script, sound, lighting), and delivered under tight performance deadlines.